

Calibrating Motion Uncertainty on Contact-Stratified Configuration Spaces

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Abstract—Reliable uncertainty representation in contact-rich manipulation requires accounting not only for the magnitude of motion uncertainty, but also for the geometry of its support. While calibrated uncertainty predictions are essential for safe and efficient planning, analytical or learned motion models are often inaccurate—due to limited data, simplifying assumptions, unmodeled effects, etc.—which can lead to unsafe executions or task failure. Contact can also change the geometry of uncertainty over future configurations: the distribution of future configurations can become multimodal or disjoint, or lie along manifolds of lower intrinsic dimension than the space of possible robot configurations. We propose Calibrated Particle-sets for Trans-dimensional Uncertainty Representation (CaPTURE), a geometry-aware, conformal prediction-based algorithm that generates probabilistically valid prediction regions of the unknown future system configuration using particle-based models of arbitrary fidelity. Our method uses a calibration dataset of system transitions to locally calibrate motion uncertainty estimates, constructing regions guaranteed to contain the future robot configuration at a user-set probability. Our calibration procedure captures how motion uncertainty varies across free-space and contact regimes, leading to the user-specified coverage within each regime. We evaluate our method on tight-tolerance peg-in-hole insertion with a manipulator. Compared with other conformal methods, CaPTURE achieves the user-specified coverage requirement both in and out of contact and a 30-percentage-point improvement in task success rate over the best baseline.

Index Terms—Geometric representations, contact-rich manipulation, stratified configuration spaces, uncertainty-aware motion planning, conformal prediction.

I. INTRODUCTION AND RELATED WORK

Deploying robots in contact-rich environments requires reasoning about how uncertainty changes across interaction regimes. For example, uncertainty about a manipulator’s end-effector location can be reduced by moving from free space into sliding contact, while impacts, sticking, and transitions between contact modes can produce abrupt changes in the distribution of future configurations. Aleatoric uncertainty arises due to measurement error, actuation delay, and surface roughness, while epistemic uncertainty arises due to model mismatch or limited task-relevant data. While prior work has demonstrated that intentional contact and compliance are important for uncertainty reduction in precise manipulation settings [1–3], it is not always trivial to induce contact safely under uncertainty. In many contact-rich tasks, violating constraints can lead to

robot or human damage. Yet, contact interactions can induce multimodal, discontinuous, and trans-dimensional uncertainty distributions over future robot configurations, which are challenging to model. There is thus a need for contact-aware, provably calibrated uncertainty estimates. Particle-based representations can capture multimodal and discontinuous motion uncertainty without assuming a parametric distribution [4, 5]. However, by themselves, these methods do not provide guarantees for quantifying errors caused by an approximate dynamics model. Similarly, while previous planning methods have exploited stratified C-spaces or manifolds of differing intrinsic dimension to represent contact-induced geometric structure [6–10], these generally assumed access to a sufficiently accurate model. Conformal prediction (CP) enables the construction of prediction regions with coverage guarantees by using an exchangeable calibration set representative of test-time conditions [11, 12]. Probabilistic conformal prediction (PCP) extends CP to particle-based models [13], while Mondrian CP partitions examples into groups and calibrates each group independently [14], enabling group-conditional coverage guarantees. Prior CP methods in robotics are either state-action independent or construct state-action-dependent Gaussian regions [15, 16]; however, these regions do not appropriately capture how uncertainty varies across contact modes, and their ellipsoids may extend into infeasible C-space regions. Building on PCP and Mondrian CP, we propose **Calibrated Particle-sets for Trans-dimensional Uncertainty Representation (CaPTURE)** for constructing calibrated next-configuration prediction regions in contact-rich settings. CaPTURE defines calibration groups jointly from the current state, action, and future C-space stratum. At inference, it evaluates candidate future contact strata and constructs the prediction region as the union of their feasible stratum-specific regions. Our contributions are: (i) a state-action-stratum-aware method for constructing feasible, trans-dimensional prediction regions from particle-based dynamics predictions of arbitrary fidelity, together with a proof that the resulting regions inherit finite-sample group-conditional coverage from Mondrian CP; and (ii) numerical coverage validation and a tight-tolerance peg-insertion evaluation under both stochastic disturbances and model mismatch. This extended abstract includes results from our full paper [17]; rollout videos are available on the project website: <https://um-arm-lab.github.io/capture>.

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II. PROBLEM STATEMENT

Consider a discrete-time stochastic system with state s_t and action a_t evolving under unknown dynamics $s_{t+1} \sim f(s_t, a_t)$. Let $c_t \in \mathcal{C}$ denote the configuration components for which we construct calibrated predictions. We assume that s_t is known. Particle filtering instead addresses state estimation from sequential observations and is complementary to our calibration of one-step motion-model error. We define the prediction input $X := (s_t, a_t)$ and output $Y := c_{t+1}$. We have access to an approximate dynamics predictor $\hat{f}$, which may be affected by aleatoric uncertainty and epistemic model mismatch. Prediction errors can cause a planner using $\hat{f}$ to select unsafe actions or inefficient trajectories. We therefore seek calibrated prediction regions over the unknown next configuration so that the planner can explicitly account for prediction errors. To obtain distribution-free, finite-sample coverage without assuming that $\hat{f}$ is accurate, we make the standard CP assumption of exchangeability:

Assumption 1: The calibration transitions $\mathcal{D}_{\text{cal}} := \{(X_i, Y_i)\}_{i=1}^n$ and the test transition (X_{n+1}, Y_{n+1}) are exchangeable.

The predictor $\hat{f}$ may be trained on an arbitrary dataset; only the calibration and test transitions must satisfy Assumption 1. For a user-specified failure rate $\alpha \in (0, 1)$, we seek a prediction region $\hat{C}(X)$ satisfying

$$\mathbb{P}\{Y_{n+1} \in \hat{C}(X_{n+1})\} \geq 1 - \alpha. \quad (1)$$

Coverage alone does not ensure that a prediction region is informative: choosing the entire C-space trivially satisfies this requirement. We therefore seek regions that adapt to the current state, action, and local contact geometry while excluding physically infeasible configurations. Let $\mathcal{C}_{\text{feas}} \subseteq \mathcal{C}$ denote the feasible C-space. We assume that it admits a finite stratification $\mathcal{C}_{\text{feas}} = \bigsqcup_{m \in M} S_m$, with a known stratum indexer $\mathcal{T} : \mathcal{C}_{\text{feas}} \rightarrow M$. We use this contact stratification as a geometric prior on the support of uncertainty over future configurations. The support of $p(Y | X)$ may span one or more strata depending on X . Our objective is to construct a calibrated prediction region $\hat{C}(X) \subseteq \mathcal{C}_{\text{feas}}$ that adapts to how uncertainty over future configurations varies with the current state, action, and future C-space stratum. These properties matter for planning: including infeasible configurations can induce unnecessary conservatism, while marginal coverage alone can conceal undercoverage in a particular contact regime.

III. METHOD: CAPTURE

Given the current state and action (s_t, a_t) , an approximate dynamics model generates predictive particles. Then, CaPTURE provides a calibrated threshold for each candidate future stratum and constructs a prediction region by taking the union of the stratum-specific regions. Figure 1 summarizes CaPTURE's planning-time prediction-region construction.

To construct prediction regions, we adopt the PCP framework from Wang et al. [13], which enables us to construct a calibrated prediction set using samples drawn from an approximate conditional distribution $\hat{f}$ over the next configurations. We

define the nonconformity score r as the distance between the true next configuration and the k_{NN} -th nearest neighbor among the L particles sampled from $\hat{f}(X_i)$. We choose the k_{NN} -th nearest-neighbor score because it provides a continuously valued proxy for the local particle density [18]. Formally, the score function is

$$r(\hat{f}(X_i), Y_i) := \text{kNN}_{k_{\text{NN}}}(Y_i, \hat{Y}_i), \quad (2)$$

where Y_i is the true next configuration, $\hat{Y}_i$ is the set of L propagated particles, and $\text{kNN}_{k_{\text{NN}}}$ denotes the distance to the k_{NN} -th closest particle. For each calibration transition (X_i, Y_i) , let $R_i = r(\hat{f}(X_i), Y_i)$ be a symmetric nonconformity score, with larger values indicating larger prediction errors. With this, we can now define the augmented calibration dataset, similarly to PCP, as $\bar{\mathcal{D}}_{\text{cal}} := \{(X_i, \mathcal{T}(Y_i), R_i) : (X_i, Y_i) \in \mathcal{D}_{\text{cal}}\}$.

We introduce the grouping function

$$g : \mathcal{X} \times M \rightarrow \{1, \dots, J\}, \quad (3)$$

which partitions examples according to the prediction-time information X and the future C-space stratum $\mathcal{T}(Y)$. Conditioning g on the future stratum allows the calibration threshold to vary across contact regimes, while conditioning on X also adapts it to state- and action-dependent prediction errors. To capture local predictive uncertainty variations in a data-driven manner, we define our g using the LOCART procedure, first presented by Cabezas et al. [19] and later adapted for dynamical systems by Marques and Berenson [15, 16]. Following LOCART, we randomly split the augmented calibration dataset into two disjoint sets $\bar{\mathcal{D}}_{\text{cal}} := \bar{\mathcal{D}}_{\text{cal}}^{\text{part}} \sqcup \bar{\mathcal{D}}_{\text{cal}}^{\text{cp}}$. Using only $\bar{\mathcal{D}}_{\text{cal}}^{\text{part}}$, we fit a regression decision tree (DTREE) with the *augmented inputs* $(X_i, \mathcal{T}(Y_i))$, where $X_i = (s_t, a_t)_i$ in our implementation, and set R_i as the target variable. The tree partitions the joint $\mathcal{X} \times M$ space to find regions of approximately constant model error. The leaves of the fitted DTREE define the grouping function $g(X, \mathcal{T}(Y))$ used by our algorithm. Within each leaf (group), we compute a separate conformal threshold $\hat{q}_j$ using only calibration points of $\bar{\mathcal{D}}_{\text{cal}}^{\text{cp}}$ that were assigned to that group. This process preserves the group-conditional coverage guarantees of Theorem 1, while producing prediction regions that adapt to the dynamics' local uncertainty structure. Importantly, g is learned independently of $\bar{\mathcal{D}}_{\text{cal}}^{\text{cp}}$ and fixed before conformal calibration. For each induced group $j \in \{1, \dots, J\}$, we follow the Mondrian procedure and define its calibration index set as $\mathcal{I}_j := \{i : (X_i, \mathcal{T}(Y_i), R_i) \in \bar{\mathcal{D}}_{\text{cal}}^{\text{cp}}, g(X_i, \mathcal{T}(Y_i)) = j\}$. Here, $\mathcal{I}_j$ contains the calibration examples assigned to group j . Mondrian CP partitions the calibration examples into groups and computes a finite-sample-corrected threshold independently within each group j :

$$\hat{q}_j := \text{Quantile} \left(\{R_i\}_{i \in \mathcal{I}_j} \cup \{\infty\}; \frac{(|\mathcal{I}_j| + 1)(1 - \alpha)}{|\mathcal{I}_j| + 1} \right). \quad (4)$$

Under Ass. 1 and a grouping function fixed independently of the data used to compute $\hat{q}_j$, the resulting prediction regions satisfy coverage of at least $1 - \alpha$ within each group [14, 20].

Theorem 1: Given Assumption 1, a grouping function g defined in Equation (3) and fixed independently of the data

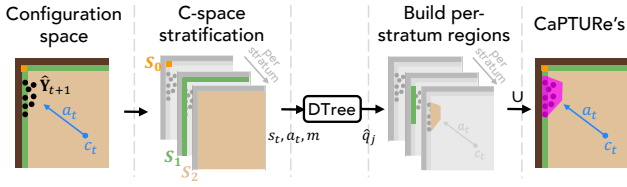


Fig. 1: **Online construction of CaPTURE.** A tree selects a calibrated threshold for each candidate future stratum; the prediction region is the union of the feasible stratum-specific regions around the predictive particles.

used to compute $\hat{q}_j$, and a symmetric score function r , let the CaPTURE prediction region be

$$\hat{C}(X_{n+1}) := \{y \in C_{\text{feas}} : r(\hat{f}(X_{n+1}), y) \leq \hat{q}_g(X_{n+1}, \mathcal{T}(y))\}.$$

For $\alpha \in (0, 1)$ and every group j with $\mathbb{P}\{g(X_{n+1}, \mathcal{T}(Y_{n+1})) = j\} > 0$, the CaPTURE prediction region $\hat{C}(X_{n+1})$ satisfies the group-conditional coverage guarantee

$$\mathbb{P}\{Y_{n+1} \in \hat{C}(X_{n+1}) \mid g(X_{n+1}, \mathcal{T}(Y_{n+1})) = j\} \geq 1 - \alpha.$$

Proof: Conditional on $\mathcal{I}_j$ and $g(X_{n+1}, \mathcal{T}(Y_{n+1})) = j$, exchangeability of the examples and symmetry of r imply exchangeability of the scores within group j [20]. The standard SplitCP argument gives coverage under this conditioning; averaging over $\mathcal{I}_j$ yields the result. ■

Since $\mathcal{T}(Y)$ is unknown at inference, we evaluate g for every candidate future stratum and take the union of the corresponding stratum-specific regions (Figure 1). Because each region is restricted to $S_m \subseteq C_{\text{feas}}$, their union excludes infeasible configurations; the group-conditional guarantees yield marginal coverage by the law of total probability.

Theorem 1 provides only single-step coverage. Formal multistep guarantees do not follow directly because later prediction regions depend recursively on earlier predicted states, particles, and regions, so exchangeability does not hold across rollout steps. We leave a multistep extension for future work.

IV. EXPERIMENTAL RESULTS

We compare our method with four relevant baselines: 1) an uncalibrated particle-based planner using $\hat{f}$ directly (*ParticleNoCP*); 2) *LUCCa* [15], a local CP method that calibrates a Gaussian $\hat{f}$ differently based on the system’s state and action, resulting in adaptive hyperellipsoidal prediction regions; 3) *PCP* [13], a particle-based conformal prediction approach that builds input-output-independent balls around each predicted particle; and 4) an ablation of our method (Section III) that does not consider the next-configuration stratum index as part of the partitioning procedure (*Ablation w/o stratum label*), i.e., the inputs to the DTtree become $X = (s_t, a_t) \in \mathcal{S} \times \mathcal{U} = \mathcal{X}$. We use $k_{\text{NN}} = 4$ and $L = 8$ in our experiments. All methods share the same approximate model $\hat{f}$ and calibration dataset D_{cal} . For methods that perform local calibration (*LUCCa*, *Ablation w/o stratum label*, *CaPTURE*), we randomly select 70% of D_{cal} for fitting the DTtree and

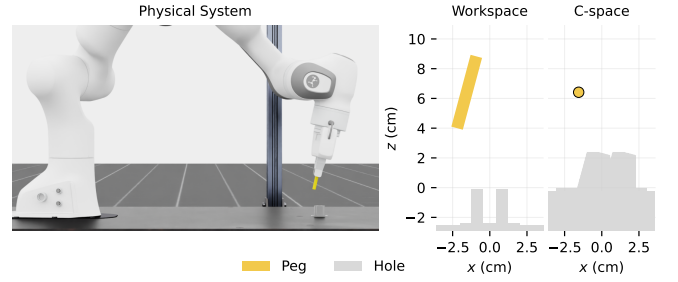


Fig. 2: **Peg-insertion task.** Left: Isaac Sim setup. Middle: planar $SE(2)$ workspace. Right: C-space slice at $\theta = -15^\circ$, where the peg cannot enter the hole.

use the remainder to obtain the per-group conformal thresholds $\hat{q}_j$, as detailed in Section III. While the particle-based methods sample from a predictive Gaussian model, *LUCCa* calibrates said $\hat{f}$ directly. We use $\alpha = 0.1$ for all experiments. We evaluate CaPTURE on a tight-tolerance peg insertion task adapted from the Factory simulation suite in Isaac Sim [21]. We control a 7-DoF manipulator (Franka Panda) to insert a cylindrical peg (diameter of 7.986 mm) into a 9.000 mm-diameter hole under both stochastic disturbances and significant model mismatch. We use a 24-mm-tall receptacle without a beveled hole, making insertion harder than in Factory’s beveled task. To facilitate contact-aware planning, we restrict the end-effector motion to the hole’s plane, reducing its possible poses from the space of three-dimensional rigid-body poses, $SE(3)$, to the space of planar rigid-body poses, $SE(2)$. Hence, the peg configuration $c_t \in SE(2)$ is parameterized by $[x, z, \theta]^\top$, where (x, z) denotes the planar position of the peg’s centroid and θ its planar orientation. The actions are desired $SE(2)$ displacements in the plane. For true rollouts, $\bar{a}_t = 1.5 \text{ clip}(a_t + \varepsilon_t, a_{\min}, a_{\max})$, with $\varepsilon_t \sim \mathcal{N}(0, \text{diag}(1.5^2, 1.5^2, 1.5^2))$. Approximate particles use noise standard deviation 0.3 and no multiplicative bias. Both use the same fixed Isaac Sim contact/friction parameters; friction mismatch is not evaluated. Renderings of the simulation environment, the workspace and the C-space are shown in Figure 2.

Due to the small peg-hole clearance, successful insertion requires precise alignment. However, near contact, even small configuration uncertainty can induce different physical interactions with the environment—the distribution over future peg configurations can become discontinuous and multimodal across contact regimes and can span both full-dimensional free space and lower-dimensional contact manifolds, producing trans-dimensional uncertainty. Additionally, overly optimistic future configuration estimation can lead to the peg getting stuck midway through insertion. This motivates the need for contact-aware and adaptive calibrated uncertainty estimates for planning.

To stratify the peg’s $SE(2)$ C-space, we heuristically label each configuration according to a “contact fingerprint.” We first decompose the rectangular cross-section of the cylindrical peg into its four vertices and four edges. To robustly detect contact between the peg and the environment, we place a disk of radius 0.75 mm around each vertex and a tube of radius 0.75 mm around each edge. Then, for each query configu-

TABLE I: Coverage (%) over 90k validation transitions.

Method	Aggregate	Free (3D)	Point (2D)	Edge (1D)
LUCCa	90.3	83.8	96.6	93.5
PCP	89.9	83.1	96.7	93.1
Ablation w/o stratum label	91.1	84.9	97.3	94.5
CaPTURE	90.4	91.0	90.1	90.3

User-set target coverage is 90%.

ration $c \in SE(2)$, we define the binary contact fingerprint $\phi(c) = (v_1, v_2, v_3, v_4, e_1, e_2, e_3, e_4) \in \{0,1\}^8$, where v_i and e_i denote the binary contact values of the i -th vertex and edge, respectively. Configurations with identical contact fingerprints are assigned to the same stratum. We discretize the C-space offline (at 0.25 mm in x and z , and 0.5° in θ) and precompute the stratum indexer $\mathcal{T}$ on this grid using the binary contact fingerprints. At planning time, grid lookup returns $m = \mathcal{T}(c_t)$, the index of the stratum S_m containing c_t . We construct D_{cal} by performing single-step rollouts with f using sampled feasible configurations and actions. We first sample initial configurations over the set of all strata in the $SE(2)$ C-space grid. For each c_t , a control input $a_t \in [a_{\min}, a_{\max}]$ is sampled uniformly. We repeat this process until we obtain $|D_{\text{cal}}| = 70,000$ transitions. Because the peg’s configuration lies in $SE(2)$, we compute distances between particles and true configurations in the Lie algebra. Given two poses $T_1, T_2 \in SE(2)$, the relative transform $T_\Delta = T_1^{-1}T_2$ has matrix logarithm $\log(T_\Delta) \in \mathfrak{se}(2)$, yielding the twist (ρ_x, ρ_z, ω) . We use the distance $\|(\rho_x, \rho_z, \ell\omega)\|_2$, where $\ell = 0.0253$ m is the peg characteristic length; for small $|\omega|$, $\ell|\omega|$ approximates the maximum rotation-induced displacement of any point on the peg. **Numerical coverage validation.** We estimate empirical coverage by evaluating 90,000 validation transitions. To compare the sizes of prediction regions across methods, we voxelize the C-space and approximate the volume of each $\hat{C}(X_{n+1})$ as the total volume of the voxels in $\hat{C}(X_{n+1})$. The mean prediction-region volume ratios relative to CaPTURE are 1.03 for LUCCa, 0.70 for PCP, and 1.09 for Ablation w/o stratum label. Table I shows the coverage results. Because the scalar stratum index supplied to the `DTree` encodes one of many binary contact fingerprints, we report in Table I a smaller, non-exhaustive set of interpretable categories: *Point Contact* refers to configurations with zero-dimensional active workspace contact support, whereas *Edge Contact* refers to configurations with nonzero-length active workspace contact support. These categories do not include all strata; e.g., cases with multiple contact points appear only in the *Aggregate* row. All methods approximately achieve marginal coverage, as expected. Although LUCCa and Ablation w/o stratum label adapt to the current configuration and action, they and PCP undercover in Free Space, likely because none of these baselines accounts for the next-configuration stratum. This suggests that explicitly representing future contact strata can improve uncertainty calibration across geometric regimes.

TABLE II: Planning over 50 Halton initial configurations.

Metric	ParticleNoCP	PCP	LUCCa	Ablation w/o stratum label	CaPTURE
Success (%) $\uparrow$	14	48	48	48	78
Avg. Steps $\downarrow$	39.9 ± 13.3	38.4 ± 13.9	41.5 ± 16.6	43.1 ± 15.9	28.7 ± 8.9

Success is insertion within 75 steps; steps are mean $\pm$ std over successful trials.

Probabilistic motion planning. We use Model Predictive Path Integral Control (MPPI) [22] for trajectory optimization in the peg-insertion task. The objective combines the Factory peg-insertion cost, a reward for overlap with goal-compatible contact configurations, and a contact-aware control penalty. MPPI uses a horizon of $H = 15$, 128 samples, discount factor $\gamma = 0.95$, and temperature $\lambda = 0.1$. As ParticleNoCP does not produce a prediction region, we compute the first-step region statistics over an axis-aligned C-space box fitted to its particles. For the peg-in-hole task we construct the calibrated prediction region only for the first step of each MPC plan. For later plan steps, we simply propagate the particles using $\hat{f}$. This is done partly for computational reasons and partly because the manipulator system is quasi-static and our CaPTURE coverage guarantees apply exclusively to single-step transitions. In quasi-static settings, one-step errors are more forgiving and can be corrected more easily in the next MPC plan. We evaluate planning performance by starting from 50 feasible initial configurations selected using a deterministic Halton sequence. Table II reports the average number of steps taken to full insertion and success rates within a 75-step (5 s) limit.

The conformal prediction-based methods outperform the ParticleNoCP baseline across the tested initial conditions. CaPTURE achieves 78% success, a 30-percentage-point improvement over each conformal baseline, and the fewest average planning steps among successful trials. Together with Table I, these results suggest that treating robot motion uncertainty as action-, configuration-, and stratum-dependent can yield informative trans-dimensional uncertainty representations for downstream probabilistic motion planning, although adequate coverage in each contact regime does not necessarily imply improved task performance.

V. CONCLUSION

We proposed a geometry-aware conformal prediction algorithm to construct probabilistically valid next-configuration prediction regions in stratified configuration spaces. Our approach combines particle-based dynamics predictors with a Mondrian-based adaptive calibration to produce uncertainty sets that can capture multimodal, discontinuous, and trans-dimensional robotic motion uncertainty and do not include infeasible regions of the robot’s configuration space. Practically, our adaptive uncertainty estimation procedure can improve task success, relative to relevant particle-based and adaptive baselines, in a tight-tolerance peg-insertion task, even under significant model mismatch and aleatoric disturbances. These results highlight contact geometry as a useful structural prior for representing motion uncertainty, rather than solely as a downstream feasibility constraint.

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